Containerization-and-DevOps

🐳 Class Task

🛠️ Tools Used

- Docker
- WSL
- Nginx
- BusyBox

⚙️ Steps

List Docker Networks

```
docker network ls
```

Inspect Default Bridge Network

```
docker network inspect bridge
```

Create Custom Bridge Network

```
docker network create my_bridge
```

Inspect Custom Network

```
docker network inspect my_bridge
```

Run First Container (Nginx) on Custom Network

```
docker run -dit --name container1 --network my_bridge nginx
```

Run Second Container (BusyBox) on Same Network

```
docker run -dit --name container2 --network my_bridge busybox
```

Test Connectivity Between Containers

```
docker exec -it container2 ping container1
```

Run Container Using Host Network

```
docker run -d --network host nginx
```

Verify Port Binding

```
ss -lntp | grep 80
```

⚠️ Notes

- Default `bridge` network is automatically created by Docker
- Custom bridge network allows container-to-container communication using names
- `ping container1` worked due to internal DNS resolution
- Host network mode shares host's networking stack
- Port `80` was directly exposed on host without mapping

✅ Outcome

- Explored Docker network types

- Created and inspected custom bridge network
- Verified container communication using ping
- Understood difference between bridge and host networking

📸 Screenshots

```
PS C:\Users\srist> wsl
wsl: Unknown key 'wsl2.systemd' in C:\Users\srist\.wslconfig:2
sristee@Sristee:/mnt/c/Users/srist$ docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
245f1befe712   bridge    bridge    local
e6aa5e1497ad   host      host      local
bb27b8c6ad33   none      null      local
sristee@Sristee:/mnt/c/Users/srist$ docker network inspect bridge
[
    {
        "Name": "bridge",
        "Id": "245f1befe712215899ac76aa0eed69e5acf7976131e0fa2c9f8998894db38c73",
        "Created": "2026-02-18T05:55:38.570532194Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.17.0.0/16",
                    "Gateway": "172.17.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {
            "com.docker.network.bridge.default_bridge": "true",
            "com.docker.network.bridge.enable_icc": "true",
            "com.docker.network.bridge.enable_ip_masquerade": "true",
            "com.docker.network.bridge.host_binding_ipv4": "0.0.0.0",
```

```
            "com.docker.network.driver.mtu": "1500"
        },
        "Labels": {}
    }
]
sristee@Sristee:/mnt/c/Users/srist$ docker network create my_bridge
370a2053131a2d3eec0706c25ebcfefb0c9e2eb006037c3a673adc2311c118ee
sristee@Sristee:/mnt/c/Users/srist$ docker network inspect my_bridge
[
    {
        "Name": "my_bridge",
        "Id": "370a2053131a2d3eec0706c25ebcfefb0c9e2eb006037c3a673adc2311c118ee",
        "Created": "2026-02-18T05:57:21.38608052Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Containers": {},
        "Options": {},
        "Labels": {}
    }
]
sristee@Sristee:/mnt/c/Users/srist$ docker run -dit --name container1 --network my_bridge nginx
```

```
sristee@Sristee:/mnt/c/Users/srist$ docker run -dit --name container1 --network my_bridge nginx
74abf512a92fe981d19d7e9434b5bcd01b42c1572a874d079331305d6b06e479
sristee@Sristee:/mnt/c/Users/srist$ ^C
sristee@Sristee:/mnt/c/Users/srist$ docker run -dit --name container2 --network my_bridge busybox
Unable to find image 'busybox:latest' locally
latest: Pulling from library/busybox
61dfb50712f5: Pull complete
Digest: sha256:b3255e7dfbcd10cb367af0d409747d511aeb66dfac98cf30e97e87e4207dd76f
Status: Downloaded newer image for busybox:latest
be9109e12c2152a812e85f1d7d08b8742d091edf83e8065aa7aa9da3667377c2
sristee@Sristee:/mnt/c/Users/srist$ docker exec -it container2 ping container1
PING container1 (172.18.0.2): 56 data bytes
64 bytes from 172.18.0.2: seq=0 ttl=64 time=0.111 ms
64 bytes from 172.18.0.2: seq=1 ttl=64 time=0.109 ms
64 bytes from 172.18.0.2: seq=2 ttl=64 time=0.291 ms
64 bytes from 172.18.0.2: seq=3 ttl=64 time=0.244 ms
64 bytes from 172.18.0.2: seq=4 ttl=64 time=0.132 ms
64 bytes from 172.18.0.2: seq=5 ttl=64 time=0.167 ms
64 bytes from 172.18.0.2: seq=6 ttl=64 time=0.143 ms
64 bytes from 172.18.0.2: seq=7 ttl=64 time=0.127 ms
64 bytes from 172.18.0.2: seq=8 ttl=64 time=0.160 ms
64 bytes from 172.18.0.2: seq=9 ttl=64 time=0.147 ms
64 bytes from 172.18.0.2: seq=10 ttl=64 time=0.200 ms
64 bytes from 172.18.0.2: seq=11 ttl=64 time=0.258 ms
64 bytes from 172.18.0.2: seq=12 ttl=64 time=0.169 ms
64 bytes from 172.18.0.2: seq=13 ttl=64 time=0.186 ms
64 bytes from 172.18.0.2: seq=14 ttl=64 time=0.150 ms
64 bytes from 172.18.0.2: seq=15 ttl=64 time=0.111 ms
64 bytes from 172.18.0.2: seq=16 ttl=64 time=0.139 ms
64 bytes from 172.18.0.2: seq=17 ttl=64 time=0.152 ms
64 bytes from 172.18.0.2: seq=18 ttl=64 time=0.187 ms
64 bytes from 172.18.0.2: seq=19 ttl=64 time=0.146 ms
64 bytes from 172.18.0.2: seq=20 ttl=64 time=0.150 ms
64 bytes from 172.18.0.2: seq=21 ttl=64 time=0.117 ms
64 bytes from 172.18.0.2: seq=22 ttl=64 time=0.146 ms
64 bytes from 172.18.0.2: seq=23 ttl=64 time=0.122 ms
64 bytes from 172.18.0.2: seq=24 ttl=64 time=0.121 ms
64 bytes from 172.18.0.2: seq=25 ttl=64 time=0.130 ms
64 bytes from 172.18.0.2: seq=26 ttl=64 time=0.135 ms
64 bytes from 172.18.0.2: seq=27 ttl=64 time=0.248 ms
```

```
64 bytes from 172.18.0.2: seq=334 ttl=64 time=0.137 ms
64 bytes from 172.18.0.2: seq=335 ttl=64 time=0.163 ms
64 bytes from 172.18.0.2: seq=336 ttl=64 time=0.229 ms
64 bytes from 172.18.0.2: seq=337 ttl=64 time=0.168 ms
64 bytes from 172.18.0.2: seq=338 ttl=64 time=0.156 ms
64 bytes from 172.18.0.2: seq=339 ttl=64 time=0.157 ms
64 bytes from 172.18.0.2: seq=340 ttl=64 time=0.240 ms
64 bytes from 172.18.0.2: seq=341 ttl=64 time=0.258 ms
64 bytes from 172.18.0.2: seq=342 ttl=64 time=0.215 ms
64 bytes from 172.18.0.2: seq=343 ttl=64 time=0.163 ms
^C
--- container1 ping statistics ---
344 packets transmitted, 344 packets received, 0% packet loss
round-trip min/avg/max = 0.088/0.188/3.029 ms
sristee@Sristee:/mnt/c/Users/srist$ docker run -d --network host nginx
f9428a8d1a33b60037f3cf62d14a565dc9959317329e031bcfe7264c593ce636
sristee@Sristee:/mnt/c/Users/srist$ ss -tulnp | grep 80
tcp   LISTEN 0      511            0.0.0.0:80         0.0.0.0:*
tcp   LISTEN 0      511               [::]:80            [::]:*
sristee@Sristee:/mnt/c/Users/srist$
```